Chemistry · Unit 0 · Level 2 (modified)

Lab Safety & Equipment



TODAY'S PLAN

- 📋 Lab rules
- 🥽 Goggles and clothes
- 🚨 Accidents
- 🧪 Lab equipment
- ☠️ Hazard symbols

Review · 5 minutes

Quick Review



🔬 Chemistry is a **hands-on lab** class.

⚠️ Safety is the **number one** rule.

❓ What safety gear have you used before?

Words to Know

Words to Know



Goggles

Lab glasses that protect your eyes. Wear them any time there are chemicals, heat, or glass.



Eyewash station

A sink that sprays water into your eyes. Rinse for 20 minutes if a chemical splashes.



Fire extinguisher

A red tank that sprays to put out a fire. Only the teacher uses it.



Beaker

A glass cup with a lip for pouring. It holds, mixes and heats liquids. The marks are only a guess.

Words to Know

More Words to Know



📏 **Graduated cylinder**: A tall, thin tube with marks to measure volume (mL). Read the bottom of the curve (meniscus).

⚖️ **Balance**: A scale that measures mass in grams (g). Press tare to set it to zero.

🧪 **Test tube**: A small, narrow glass tube for holding, mixing or heating small amounts. Hold it with a test tube holder.

☠️ **Hazard symbol**: A label picture that tells the danger: flammable, corrosive, toxic, irritant, oxidizing, biohazard.

Steps

Lab Rules from the Safety Contract



1. 👂 Follow all directions. Ask if you are not sure.
2. 🖐️ Do not touch equipment or chemicals until the teacher says.
3. 🚫 No food, drinks or gum. Never drink from lab glass.
4. 🤷 No fooling around. Stay at your lab station.
5. 🧼 Clean up and wash your hands when you finish.

🃏 Check 1 of 5



When can you touch the lab equipment?

Hold up 1, 2, or 3:

1️⃣ When you walk in

2️⃣ When the teacher says

3️⃣ Any time

✅ Answer · Card 1 of 5



When the teacher says

Rule 4: do not touch anything until you are told to.

Learn

Goggles and Clothes

🥽 Wear **goggles** any time there are **chemicals, heat or glass**. No exceptions.

💇 Tie back **long hair**. Tuck in loose clothes and jewelry.

👟 Wear **closed-toe shoes**. No sandals.

🥼 Wear a **lab apron** during labs.



🃏 Check 2 of 5



Which shoes are safe in the lab?

Hold up 1, 2, or 3:

1️⃣ Sandals

2️⃣ Flip-flops

3️⃣ Closed-toe shoes

✅ Answer · Card 2 of 5



Closed-toe shoes

Shoes must cover your whole foot.

Learn

Accidents and Injuries

📢 Tell the teacher about **any** spill, break, cut or burn, even small ones.

🗣️ If someone is hurt, yell **“Code one, Code one!”**

🚿 Chemical in eyes or on skin? Rinse at the **eyewash or safety shower** for **20 minutes**.

🧹 Never pick up **broken glass** with your hands.



🃏 Check 3 of 5



A chemical splashes in your eye. How long do you rinse?

Hold up 1, 2, or 3:

1️⃣ 20 minutes

2️⃣ 10 seconds

3️⃣ 1 minute

✅ Answer · Card 3 of 5

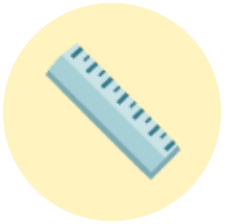


20 minutes

Rinse for at least 20 minutes, then tell the teacher.

Learn

Measuring Tools (L0.3)



Graduated cylinder

volume in mL; read the bottom of the meniscus



Balance

mass in grams; press tare to zero



Thermometer

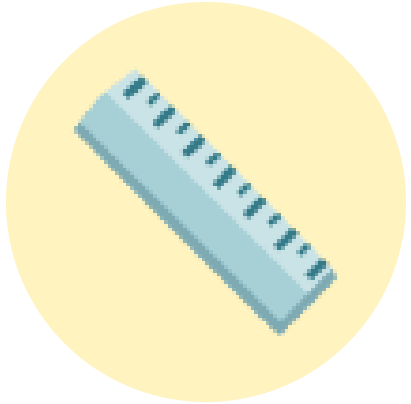
temperature in °C



Pipet

moves small drops of liquid

🃏 Check 4 of 5



Which tool measures the volume of a liquid best?

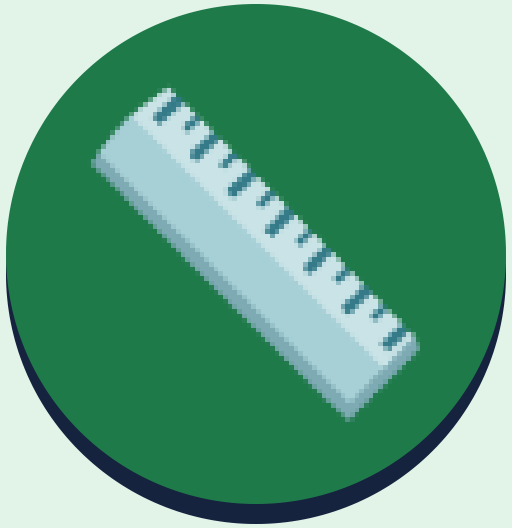
Hold up 1, 2, or 3:

1️⃣ Beaker

2️⃣ Balance

3️⃣ Graduated cylinder

✅ Answer · Card 4 of 5



Graduated cylinder

A beaker's marks are only a guess. A graduated cylinder is exact.

Learn

Hazard Symbols (L0.3 Station 6)



Flammable
can catch fire



Corrosive
destroys skin or metal



Toxic
poison



Irritant
makes skin red or itchy

🃏 Check 5 of 5



A label says CORROSIVE. It can…

Hold up 1, 2, or 3:

1️⃣ Destroy skin or metal

2️⃣ Catch fire

3️⃣ Make you sleepy

✅ Answer · Card 5 of 5



Destroy skin or metal

Corrosive chemicals eat away at skin and metal.

Wrap-Up

Recap



Rules

listen, no food, stay at your station



Goggles

chemicals, heat or glass



Accidents

tell, Code one, rinse 20 min



Equipment

right tool for the job

Wrap-Up

Exit Ticket



1️⃣ Name 2 times you must wear goggles.

2️⃣ What do you do if a chemical splashes in your eye?

3️⃣ Which tool measures mass? Which measures volume?